

SOLID-STATE VOLTAGE STABILISER



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In India, we have a large power distribution system with heavy distribution losses and variations in industrial/domestic load. This results in voltage variations that may damage electrical/electronic appliances like light, fan, television, mixer-grinder, air-conditioner, heater, water pump, toaster, etc.

Here, we describe how to make a solid-state voltage stabiliser that does not use electromechanical relays and is suitable for most purposes. Key features of the solid-state voltage stabiliser are:

1. Wide range of voltage variation from 120 V to 280 V
2. Only two settings are required – low voltage and high voltage
3. Stabilised output of 220V
4. Compact size
5. Silent operation and no relay chattering sound
6. Bar graph LED voltage indicator

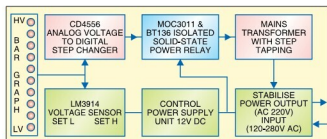


Fig. 1: Block diagram of solid-state voltage stabiliser

7. Low/high voltage indicator and cut-off protection

The block diagram of a solid-state voltage stabiliser is shown in Fig. 1. The circuit diagram comprises following four sections:

1. Analogue voltage to digital step changer
2. Isolated solid-state power relay
3. Control power supply unit
4. Mains transformer

Analogue voltage to digital step changer. The circuit diagram of a solid-state voltage stabiliser is shown in Fig. 2. The heart of the stabiliser is IC1

(LM3914) bar display driver. It is used as LED-type bar graph voltmeter with lower voltage and upper voltage settings through presets VR1 and VR2. IC1 senses mains voltage. The difference

between the lower voltage and upper voltage is divided into 10 steps. Every LED indicates one step or one voltage level and is lit depending on the level of voltage received.

All the 10 outputs of IC1 that are used to lit the LEDs are also fed as inputs to dual decoder/demultiplexer CD4556. CD4556 is used for converting analogue voltage to digital steps to ensure that, at a given time, only one tapping of mains transformer gets input supply voltage from mains. In all conditions only one step can be active as per analogue input voltage.

Output of LM3914 with Input to CD4556

	Pin No.	1	2	3	15	14	13	1	2	3	15	14	13	1	2	3	Output
	CD4556	IC2(A)			IC2(B)			IC3(A)			IC3(B)			IC4(A)			
	Step	En	A0	A1	En	A0	A1	En	A0	A1	En	A0	A1	En	A0	A1	
Low V	0	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	Dis All Low V
	1	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	En IC2(A), Q-3
	2	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	Q2
	3	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	En IC2(B), Q-3
	4	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	Q2
	5	L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	En IC3(A) Q-3
	6	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H	Q2
	7	L	L	L	L	L	L	L	L	L	L	H	H	H	H	H	Q-0, IC3(B) Q-3
	8	L	L	L	L	L	L	L	L	L	L	L	H	H	H	H	Q2
High V	9	L	L	L	L	L	L	L	L	L	L	L	L	L	H	H	En IC4(A) Q-3
	10	L	L	L	L	L	L	L	L	L	L	L	L	L	L	H	Q2
	Pin No.	1	18	17	16	15	14	13	12	11	10	Output of IC-1 LM3914					

$$\text{Transformer Taping} = \frac{\text{High voltage} - \text{Low voltage}}{\text{Step control}}$$